

Innovation or Imitation? The effect of spillovers and competitive pressure on firms' R&D strategy choice

Olga Slivko and Bernd Theilen*[†]

Departament d'Economia and CREIP, Universitat Rovira i Virgili

January, 2012

Abstract

In this paper a firm's R&D strategy is assumed to be endogenous and allowed to depend on both internal firms' characteristics and external factors. Firms choose between two strategies, either they engage in R&D or abstain from own R&D and imitate the outcomes of innovators. This yields three types of equilibria in which either all firms innovate, some firms innovate and others imitate, or no firm innovates. Firms' equilibrium strategies crucially depend on external factors. We find that the efficiency of intellectual property rights protection positively affects firms' incentives to engage in R&D, while excessive competitive pressure has a negative effect. In addition, smaller firms are found to be more likely to become imitators when the product is homogeneous and the level of spillovers is high. Regarding social welfare our results indicate that strengthening intellectual property protection can have an ambiguous effect. In markets characterized by a high rate of innovation a reduction of intellectual property rights protection can discourage innovative performance substantially. However, a reduction of patent protection can also increase social welfare because it may induce imitation. This indicates that policy issues such as the optimal length and breadth of patent protection cannot be resolved without taking into account specific market and firm characteristics.

Journal of Economic Literature Classification Numbers: C35, D43, L13, L22, O31.

Keywords: Innovation; imitation; spillovers; product differentiation; market competition; intellectual property rights protection.

*Correspond to: Departament d'Economia, Universitat Rovira i Virgili, Avinguda de la Universitat 1, E-43204 Reus, Spain, E-mail address: olga.slivko@urv.cat, bernd.theilen@urv.cat.

[†]An earlier version of this paper has been presented at the "European Network on Industrial Policy" workshop in Florence, the XVI edition of "Dynamics, Economic Growth, and International Trade" conference in Saint-Petersburg and the XXVI edition of the "Jornadas de Economía Industrial" in Valencia and seminars in Universitat Rovira i Virgili and Universitat de les Illes Balears. We are very grateful to the participants in these meetings and especially to Xosé-Luís Varela, Matthias Dahm, Ricardo Flores-Fillol, Susana Iranzo, Galina Zudenkova for their comments and suggestions. The authors also acknowledge financial support from the Spanish "Ministerio de Ciencia e Innovación" under project ECO2010-17113.

1 Introduction

The economic literature on innovation has provided two confronting views concerning the relationship between innovation and imitation. According to the Schumpeterian view, imitation dampens innovation as it renders innovative efforts unprofitable. In this view, intellectual property rights (IPR) protection is a necessary mechanism that provides incentives for firms to engage in R&D and encourages technology transfer between firms. Therefore, a strong protection of intellectual property rights would be the optimal R&D policy (Arora and Gambardella, 1994; Gallini and Scotchmer, 2002; Gans and Stern, 2003; Gans et al., 2008). This view has recently been challenged by Aghion et al. (2001), Bessen and Maskin (2009) and Zhou (2009) who show that stronger imitation fosters innovative efforts by incumbent firms. So, IPR protection can block the future development of technologies. Looking at these contradictory views the question of what should be the optimal balance of innovation and imitation arises. Certainly, the evidence on innovative activity at the firm level suggests elevated heterogeneity in innovative performance within as well as across markets. The heterogeneity observed is the result of firms' decisions to engage in R&D or to abstain from own R&D and imitate the outcomes of innovators. This indicates that any policy intervention might not only affect the level of a firm's R&D performance but also the strategies adopted by firms. In this paper we develop a model that allows us to analyze how external market parameters such as the intensity of IPR protection, market competition, or the degree of product differentiation affect firms' R&D strategy choices.

Theoretical studies have analyzed the effect of possible imitation on innovative incentives in two frameworks, economic growth models (Grossman & Helpman, 1991; Aghion & Howitt, 1992) and oligopolistic competition models (Zhou, 2009). However, most of this literature assumes that innovators and imitators are exogenously given. Exceptions are Segestrom (1991) and Amir and Wooders (2000). Applying an economic growth model, Segestrom (1991) allows firms to participate in innovative or imitative R&D races. In the steady-state, firms' equilibrium R&D strategies depend on the distribution of previous R&D outcomes and the relative price of imitation. Firms are found to benefit more from imitation in industries with a single leader, while in industries with several leaders innovation is a more profitable strategy. In a standard oligopoly framework, Amir and Wooders (2000) show that, in equilibrium, firms choose their R&D strategies asymmetrically which gives rise to an innovator/imitator configuration in the market. Surprisingly, until now no paper has analyzed the welfare and policy implications of alternative market structures on innovation incentives when firms endogenously choose their R&D strategy.¹ In this paper we develop a theoretical model that explains how firms endogenously choose their R&D strategy between innovation and imitation and analyze how market characteristics determine firms' R&D strategy choices and the welfare implications of alternative IPR protection regimes. Contrary to Amir and Wooders (2000), spillovers from innovators to imitators are modelled as in d'Aspremont and Jacquemin (1988) and not in an all-or-nothing probabilistic fashion. Furthermore, we allow for ex-ante asymmetric firms and extend the analysis to markets with more than two firms. Our analysis derives empirically testable predictions and appoints to some changes in IPR protection policies.

Our set-up is a two-stage Cournot model with differentiated products and strategic R&D choice. In stage 1, firms decide simultaneously what R&D strategy to apply, innovation or imitation. In stage 2, firms compete in quantities with differentiated products, conditional on their R&D strategy choice. We characterize the equilibria of this game and show how different innovation patterns that depend on the extent of spillovers, asymmetries between firms and competitive pressure arise. Three types of equilibria are obtained: equilibria in which all firms innovate, equilibria in which firms choose asymmetric R&D strategies, and equilibria in which no firm innovates. We find that the efficiency of IPR protection positively affects firms' incentives to engage in R&D, while competitive pressure has a negative effect. In addition, smaller firms are found to be more likely to

¹Segestrom (1991) assumes a unique market structure for his R&D innovation and imitation races. Amir and Wooders (2000) allow for different market structures but their welfare implications are focused on research joint ventures.

become imitators when products are homogenous and the level of spillovers is high. Regarding social welfare, our results indicate that the strengthening of IPR protection can have an ambiguous effect. If a market is characterized by a high rate of innovation, a reduction of IPR protection can discourage innovative performance substantially. However, a reduction of IPR protection can also increase social welfare because it may induce imitation. Furthermore, regarding the welfare implications of market dominance we find that a positive effect on social welfare is obtained when market dominance means that the larger firm becomes an innovator or when it does not affect the firms' R&D strategies. However, when more market dominance causes the smaller firm to abstain from innovation, the welfare effect is negative. This happens when products are highly differentiated and spillovers are low.

The main policy implication derived from our analysis is that a common IPR protection policy for all markets might be inappropriate. This is because a policy that is beneficial for a certain type of market might discourage innovation and technological progress in another with different characteristics. The analysis of spillovers on social welfare shows that a reduction of IPR protection, intended to induce imitation, can discourage innovative performance substantially in markets that are characterized by a high rate of innovation. Then, an additional reduction of IPR protection induces more imitation and increases welfare. However, after a certain point, the reduction of patent protection completely discourages innovation and therefore reduces social welfare. Moreover, an IPR protection policy must be tightly coordinated with the competition policy. This is because external parameters such as IPR protection and competitive pressure jointly affect the firms' R&D strategy choice.

This paper is related to a large literature on the relationship between market structure and innovation strategy. Specifically, it is related to two strands in the literature. The first strand analyzes how firms' R&D investments are affected by market competition. Pioneer works in this field are those of Schumpeter (1934 and 1942) who argues that, on the one hand, market pressure may foster firms' innovation. But, on the other hand, market pressure may also decrease firms' R&D investments because monopoly power of larger firms acts as a major accelerator of technological progress. Actually, there is still no accordance on this Schumpeterian debate in theoretical and empirical studies. For example, some authors argue that more intensive market competition decreases a firm's incentives for innovation because when advantages from innovation are temporary, only sufficient market power guarantees that firms invest in R&D (Arrow, 1962; Futia, 1980; Gilbert and Newbery, 1982; Reinganum, 1983; or Zhou, 2009). This argument is supported by empirical studies that find that market concentration increases the pace of innovative change. For instance, Henderson and Cockburn (1996) show that large firms in the US pharmaceutical industry perform R&D more efficiently, as they can enjoy scale and scope economies. Using patent data of UK manufacturing firms, Cefis (2003) finds that, due to innovative effort, the contribution of large firms to aggregated industrial performance is above the industry mean. On the other hand, market concentration is also argued to have a dampening effect on innovation because more intensive competition acts as an important incentive for firms to innovate (Dasgupta and Stiglitz, 1980). Again, this theoretical argument is supported by empirical evidence (Geroski, 1990; Blundell et al., 1999). These contradictory results led to the hypothesis that the effect of market competition on firms' innovative efforts is non-monotonic. For example, Boone (2000) finds that when competition is weak, the incentives of less efficient firms to innovate increase. However, when competition becomes more intense, the incentives of efficient firms to innovate grow. Aghion et al. (2005) suggest the existence of an inverted-U relationship. Both, a low or high level of competition provide low incentives to innovate while a medium level of competition fosters innovation of firms operating on a similar technological level ("neck-and-neck firms"). On the contrary, Tishler and Milstein (2009) find that R&D investments decrease with competitive pressure. However, at a certain level of competition firms engage in "R&D wars" and spend excessively on R&D. Our results also indicate a non-monotonic relationship and highlight the importance of factors such as IPR protection and product differentiation for the effect of changes in competitive pressure on innovation.

The second strand of the literature to which this paper is related are studies that, contrary to the above

literature that assume that all firms are innovators, allow for heterogeneity in firms' R&D strategies by distinguishing between firms that innovate and those that imitate innovators. As empirical evidence suggests, most markets are characterized by an elevated heterogeneity of R&D activities. So, in most markets we find a core of firms that are persistent innovators while other firms either are occasional innovators or imitators (Cefis and Orsenigo, 2001; Cefis, 2003). Czarnitzki et al. (2008) find that, depending on a firm's role in the market, competitive pressure might have a different effect on innovative effort. For example, while entry pressure decreases the average investment per firm, it increases innovative effort of market leaders. In theoretical studies (Grossman and Helpman, 1991; Aghion and Howitt, 1992; Zhou, 2009), imitation is shown to foster the innovation activity of technological leaders. This finding challenges the common view that patent protection should be strengthened. In fact, some studies argue that strong IPR protection may slow down the development of countries and decrease world welfare and consumer surplus (Helpman, 1993; Bessen and Maskin 2009; Che et al., 2009; Fershtman and Markovich, 2010). Additionally, Braguinsky et al. (2007) find that the relationship between innovation and imitation itself depends on other factors such as the maturity of an industry. When the industry is young and small, innovators do not have incentives to prevent imitation. But when the industry expands, innovative effort decreases because of imitation pressure. The results in this paper contribute to this literature first, by showing how different market characteristics give rise to different innovator/imitator configurations. Second, the results show that an increase of IPR protection can both increase and decrease social welfare depending on the underlying market characteristics. This result reconciliates the confronting views on the role of patent protection strength.

The rest of the paper is arranged as follows. Section 2 presents a theoretical Cournot duopoly model of R&D strategy choice. Section 3 includes extensions for the cases of asymmetric firms and more than two-firm competition. Section 4 discusses the policy implications of our findings. Proofs are in the Appendix.

2 A duopoly model

In this section we develop a two-stage Cournot duopoly model with differentiated products and strategic R&D choice. In stage 1, firms decide simultaneously what R&D strategy to apply, innovation or imitation. In stage 2, firms compete in quantities with differentiated products, conditional on their R&D strategy choice. We assume that each firm produces a single good and that the two goods are substitutes. The inverse demand function of good i is:

$$p_i = a - bq_i - dq_j, \quad i, j = 1, 2, i \neq j, \quad (1)$$

where p_i is the price and q_i is the quantity of good i . We assume that $a > 0$, $b > 0$, $d \geq 0$. Furthermore, the absolute value of the own-price effect on the quantity demanded is assumed to be higher than the corresponding effect of the price of the substitute, thus $b - d \geq 0$.

The R&D strategy at stage 1 is realized by the choice of a binary variable x_i , where $x_i = 1$ stands for the firm's decision to engage in R&D and $x_i = 0$ means that the firm abstains from innovation. R&D investment allows a firm to reduce its unit production cost c by the amount γx_i at cost Kx_i , where $\gamma \in [0, 1]$ and $K > 0$ are known constants. However, if a firm abstains from investing in R&D at stage 1, due to spillovers, its production cost still is reduced by imitating the rival's R&D outcomes.² Concretely, if firm i innovates and firm j abstains from innovation but decides to imitate, the unit cost reduction for firm j is $\sigma\gamma x_i$. The parameter σ indicates to what extent a cost reduction of firm i allows firm j to reduce its own production costs. We assume that $\sigma \in [0, 1]$, where $\sigma = 0$ indicates that there are no spillovers and $\sigma = 1$ means that firm j obtains the same cost

²Notice, that the assumption that imitation is costless is not essential for the results. With costly imitation we could interpret K as the difference between innovation and imitation costs.

saving as firm i without any additional investment.³ Resuming this, the unit production cost of firm i is given by:

$$c_i(x_1, x_2) = \begin{cases} c - \gamma x_i & \text{for } x_1 = 1 \text{ and } x_2 = 1 \\ c - \gamma(x_i + \sigma x_j) & \text{else} \end{cases}, \quad i, j = 1, 2, i \neq j \quad (2)$$

where $c > \gamma$. The innovation activity analyzed in this paper is cost-reducing (process innovation). However, the results can be straightforwardly generalized to the case of product innovation.⁴

Total production costs are $C_i(x_1, x_2) = c_i(x_1, x_2) q_i$. The objective of firm i is to choose the R&D strategy that maximizes profits:

$$\Pi_i(x_1, x_2) = \pi_i(x_1, x_2) - K x_i \quad (3)$$

where $\pi_i(x_1, x_2)$ denotes operating profits obtained in stage 2.

The solution concept is Subgame Perfect Nash Equilibrium (SPNE) and the game is solved by backward induction. First, for given R&D strategies the optimal equilibrium outputs are solved in the second stage. Then, firms' profit-maximizing R&D strategies in stage 1 are derived.

In *stage 2*, firm i chooses the output q_i in order to maximize its operating profit:

$$\pi_i = (a - b q_i - d q_j) q_i - c_i q_i, \quad i, j = 1, 2, i \neq j. \quad (4)$$

From straightforward calculations we find that the Nash-Cournot equilibrium output for firm i is given by:

$$q_i = \frac{2b(a - c_i) - d(a - c_j)}{4b^2 - d^2}, \quad i, j = 1, 2. \quad (5)$$

Notice, that the output of firm i is positive as long as $(a - c_i) > \frac{d}{2b}(a - c_j)$. Firm i 's optimal equilibrium operating profit is given by $\pi_i = b q_i^2$, $i = 1, 2$.

In *stage 1*, firms choose their profit-maximizing R&D strategy. When both firms engage in R&D (i.e. $x_1 = x_2 = 1$), quantities and profits are:

$$q_i(1, 1) = \frac{a - c + \gamma}{2b + d} \quad \text{and} \quad \Pi_i(1, 1) = b q_i(1, 1)^2 - K. \quad (6)$$

If none of the firms engages in R&D (i.e. $x_1 = x_2 = 0$), quantities and profits are equal to those of the classical Cournot model with differentiated products:

$$q_i(0, 0) = \frac{a - c}{2b + d} \quad \text{and} \quad \Pi_i(0, 0) = b q_i(0, 0)^2. \quad (7)$$

Finally, if one firm engages in R&D, say firm 1, and firm 2 decides to imitate, the corresponding quantities and profits are given by:⁵

$$q_1(1, 0, \sigma) = \frac{(2b - d)(a - c) + (2b - d\sigma)\gamma}{4b^2 - d^2} \quad \text{and} \quad \Pi_1(1, 0, \sigma) = b q_1(1, 0, \sigma)^2 - K \quad (8)$$

$$q_2(1, 0, \sigma) = \frac{(2b - d)(a - c) + (2b\sigma - d)\gamma}{4b^2 - d^2} \quad \text{and} \quad \Pi_2(1, 0, \sigma) = b q_2(1, 0, \sigma)^2. \quad (9)$$

³Here we focus on asymmetric spillovers (from the innovator to the imitator), which is justified when firms perform different roles in the market (see Bower and Christensen, 1995; De Bondt, 1996; Amir and Wooders, 2000).

⁴See Tishler and Milstein (2009) for this.

⁵Further on, without loss of generality, we assume that firm 1 innovates and firm 2 imitates.

The equilibrium R&D strategies are obtained as a result of each firm's best strategic response to the profit-maximizing strategy of the rival. The most interesting parameters that affect a firm's R&D strategy choice are the extent of spillovers in the industry and the degree of product differentiation, which sometimes is interpreted as a measure for the intensity of competition in the industry.⁶

The value of the spillover parameter reflects the legal and technical framework of the industry, specifically, the level of IPR protection or the ease of knowledge transfer in the market. The polar cases are a blue print diffusion in the absence of IPR protection, or the absolute ease of replication ($\sigma = 1$), and an absence of any knowledge diffusion when an invention can be completely protected by a patent, or a high level of knowledge sophistication that makes it impossible to replicate ($\sigma = 0$). In practice, most markets can be characterized by some intermediate level of spillovers. The degree of product differentiation varies from completely different products ($d = 0$) to homogeneous or identical products ($d = 1$). To exclude trivial cases we make restrictions on R&D costs:

Assumption 1. Let $\underline{K} < K < \bar{K}$ where $\underline{K}$ is defined by $\Pi_i(1, 1) = \Pi_i(0, 0)$, $i = 1, 2$ and $\bar{K}$ is defined by $\Pi_2(1, 1) = \Pi_2(1, 0, 0)$.

This assumption guarantees that costs are not too low such that making no R&D is a possible choice and that costs are not too high such that in the absence of spillovers firms are interested in investing in R&D. Thus, the focus of the analysis is to characterize the conditions under which engaging in or abstaining from own R&D is a non-trivial Nash equilibrium. Assuming that firm 1 decides to innovate, from expressions (6) and (9) we see that firm 2 faces a trade-off when choosing between innovation and imitation. On the one hand, if firm 2 decides to innovate it must pay a cost K , which in turn allows to obtain a reduction of unit production costs. On the other hand, if firm 2 decides to imitate, it saves the payment of the R&D cost K . Then, however, the decrease in unit production costs will be lower and depend on the R&D outcome of the innovator and the value of the spillover parameter.

To characterize the equilibria of the two-stage game, let $\underline{\sigma}$ be implicitly defined by $\Pi_i(1, 1) - \Pi_2(1, 0, \underline{\sigma}) = 0$ and $\bar{\sigma}$ by $\Pi_1(1, 0, \bar{\sigma}) - \Pi_i(0, 0) = 0$. We obtain the following proposition:

Proposition 1 (*Existence of equilibria*)

For given parameter values (d, b, a, c, γ) the equilibrium R&D strategies are characterized as follows:

- (i) When spillovers are low ($\sigma \leq \underline{\sigma}$) there exists a pure strategy SPNE, in which both firms engage in R&D (Region I).
- (ii) When spillovers are intermediate ($\underline{\sigma} \leq \sigma \leq \bar{\sigma}$) there exist multiple pure strategy SPNE, in which one firm engages in R&D and the other firm chooses to imitate (Region II).
- (iii) When spillovers are high ($\bar{\sigma} \leq \sigma$) there exists a pure strategy SPNE, in which none of the firms engages in R&D (Region III).

Furthermore, $\partial \bar{\sigma} / \partial K < 0$, $\partial \underline{\sigma} / \partial K < 0$, $\partial \bar{\sigma} / \partial \gamma < 0$, $\partial \underline{\sigma} / \partial \gamma < 0$, $\partial \bar{\sigma} / \partial (a - c) > 0$ and $\partial \underline{\sigma} / \partial (a - c) > 0$.

The three regions are displayed in Figure 1. In Region I there exists a unique SPNE in pure strategies in which both firms innovate. This equilibrium is obtained when spillovers and R&D costs are low and when markets are large. Actually, Region I corresponds to the case of a highly innovative competitive industry with

⁶ Among others, this interpretation is used by Tishler and Milstein (2009). However, as discussed by Theilen (2012) and from the results of Vives (2008) this interpretation should be taken with some care.

either an elevated level of knowledge protection or knowledge sophistication such that innovations are difficult to copy. In Region III there exists a unique SPNE in which none of the firms innovates though innovation would be individually profitable. This equilibrium emerges in the presence of high spillovers and elevated product homogeneity. Region III is an example of markets where competition together with free knowledge flows discourages innovation.

While firms' R&D strategies in Regions I and III are symmetric, in Region II both firms choose opposed strategies in equilibrium. Furthermore, we have multiple equilibria with one innovating and one imitating firm.⁷ This is the case for intermediate spillover levels. An increase in R&D cost K and a decrease in market size $a - c$ shifts the curves to the "south-west" so that Region I becomes smaller and innovation in Region II holds for lower spillover level. Amir and Wooders (2000) also find that initially symmetric firms apply different R&D strategies in equilibrium and therefore perform asymmetrically. However, in our model this is not necessarily the case but depends on both the intensity of competition and the degree of IPR protection. We deal with the multiplicity of equilibria in Region II by assuming that either pure-strategy equilibrium is played with equal probability. The qualitative nature of the results does not depend on the selection of the equilibrium but reflects the initial symmetry between firms and their choices. So, if we allow for mixed strategy equilibria the comparison of payoffs and social welfare between regions remains the same.

Proposition 2 (*The effect of σ on aggregated output and social welfare*)

- (i) Output and welfare are lower in the area of high spillovers (in Region III) than in the area of low spillovers (in Region I);
- (ii) Output is increasing in σ and welfare is convex in σ for intermediate spillovers (in Region II);
- (iii) Output decreases when passing from low to intermediate and from intermediate to high spillovers. Welfare can increase or decrease when passing from low to intermediate spillovers (from Region I to Region II) and welfare decreases when passing from intermediate to high spillovers (from Region II to Region III).

To illustrate the results of Proposition 2, we display the effect of changes in spillovers on aggregate industry output and welfare for some parameter values in Figure 2. We obtain two principal results. First, the relationship between the level of spillovers and aggregated industry output is non-monotonic. So, since the industry output is lower when spillovers are high, for intermediate spillover levels an increase of σ increases industry output. Second, a similar result holds for the relationship between the level of spillovers and social welfare with the difference that welfare might be even higher for an intermediate spillover level than for a low one.

These results imply that the answer to the question of whether spillovers favor or discourage innovation is not straightforward. In our model spillovers have two different effects on the level of R&D output. A first effect is that with higher spillovers, in equilibrium, fewer firms are innovators. This decreases R&D output. A second effect is that with higher spillovers imitators obtain greater efficiency gains from the use of innovators' less costly technology. This increases aggregated industry output and social welfare. While the first effect tends to dominate if changes in spillover levels are large, the second effect dominates for small variations of σ . However, because of discontinuities, small changes in spillovers can also lead to important reductions in R&D output, aggregated industry output and social welfare. Therefore, a crucial question is to find the right level of

⁷See also Zhou (2009) who assumes exogenously one innovating firm and n imitators and analyzes how competitive pressure affects the innovator's incentives to engage in R&D.

spillovers. This result provides a possible explanation to the long and controversial discussion concerning the duration of patents.⁸

Regarding the effect of product differentiation on aggregated industry output and social welfare we find that when products are more homogenous firms need more IPR protection in order to maintain incentives for innovation. This finding is supported by empirical evidence for U.S. drug companies in the 1970s and 1980s. For this data, Shankar et al. (1998) show that the capacity to differentiate products acts as an important factor for firms' survival.

3 Extensions

3.1 Asymmetric firms

The results of Section 2 can be extended for the case of initially asymmetric firms where the inverse demand function of good i is:

$$p_i = a_i - bq_i - dq_j, \quad i, j = 1, 2, i \neq j, b - d \geq 0, \quad (10)$$

and the unit production cost of firm i is given by:

$$c_i(x_i, x_j) = \begin{cases} c_i - \gamma x_i & \text{for } x_i = 1 \text{ and } x_j = 1 \\ c_i - \gamma(x_i + \sigma x_j) & \text{else} \end{cases}, \quad i, j = 1, 2, i \neq j. \quad (11)$$

Without loss of generality we assume that initially firm 1 is larger than firm 2, $a_1 - c_1 > a_2 - c_2$. Defining $M = (a_1 - c_1) + (a_2 - c_2)$, and $\epsilon = (a_1 - c_1)/M$, this means that $\epsilon \in (\frac{1}{2}, 1)$. The Nash-Cournot equilibrium output for firm i is given by:

$$q_i = \frac{2b(a_i - c_i) - d(a_j - c_j)}{4b^2 - d^2}, \quad i, j = 1, 2. \quad (12)$$

Now, four possible situations may occur. When none of the firms innovates, firms' outputs are given by:

$$q_1(0, 0) = \frac{2b\epsilon M - d(1 - \epsilon)M}{4b^2 - d^2} \quad \text{and} \quad q_2(0, 0) = \frac{2b(1 - \epsilon)M - d\epsilon M}{4b^2 - d^2}. \quad (13)$$

The corresponding profits are $\Pi_1(0, 0) = bq_1(0, 0)^2$ and $\Pi_2(0, 0) = bq_2(0, 0)^2$. When both firms innovate, the output of each firm is:

$$q_1(1, 1) = \frac{2b\epsilon M - d(1 - \epsilon)M + (2b - d)\gamma}{4b^2 - d^2} \quad \text{and} \quad q_2(1, 1) = \frac{2b(1 - \epsilon)M - d\epsilon M + (2b - d)\gamma}{4b^2 - d^2}. \quad (14)$$

The corresponding profits are $\Pi_1(1, 1) = bq_1(1, 1)^2 - K$ and $\Pi_2(1, 1) = bq_2(1, 1)^2 - K$. When only firm 1 engages in R&D and firm 2 decides to imitate, the firms' outputs are:

$$q_1(1, 0, \sigma) = \frac{2b\epsilon M - d(1 - \epsilon)M + (2b - d\sigma)\gamma}{4b^2 - d^2} \quad \text{and} \quad q_2(1, 0, \sigma) = \frac{2b(1 - \epsilon)M - d\epsilon M + (2b\sigma - d)\gamma}{4b^2 - d^2}. \quad (15)$$

⁸ Helpman (1993), Aghion et al. (2001), Bessen and Maskin (2009) and Zhou (2009), for example, argue against patents because of their redundant and excessive protection, which discourages firms' incentives for innovation. Halmenschlager (2006) and Fershtman and Markovich (2010) also find that the presence of patent protection on an intermediate stage would delay the pace of innovation and that lower spillovers are not the optimal public policy. Finally, Boldrin and Levine (2008) find that the greater the market scale (industry size) the more reduced should be IP protection. On the other hand, Arora and Gambardella (1994), Gans and Stern (2003), Gans et al. (2008) argue that IPR protection is essential for the existence of a market for technology.

The firms' profits are $\Pi_1(1, 0, \sigma) = bq_1(1, 0, \sigma)^2 - K$ and $\Pi_2(1, 0, \sigma) = bq_2(1, 0, \sigma)^2$. Finally, if firm 2 engages in R&D and firm 1 decides to imitate, the firms' outputs are:

$$q_1(0, 1, \sigma) = \frac{2b\epsilon M - d(1 - \epsilon)M + (2b\sigma - d)\gamma}{4b^2 - d^2} \quad \text{and} \quad q_2(0, 1, \sigma) = \frac{2b(1 - \epsilon)M - d\epsilon M + (2b - d\sigma)\gamma}{4b^2 - d^2}. \quad (16)$$

The firms' profits are $\Pi_1(0, 1, \sigma) = bq_1(0, 1, \sigma)^2$ and $\Pi_2(0, 1, \sigma) = bq_2(0, 1, \sigma)^2 - K$.

Let $\underline{\sigma}_1$ be implicitly defined by $\Pi_1(1, 1) = \Pi_1(1, 0, \underline{\sigma}_1)$, $\bar{\sigma}_1$ by $\Pi_1(0, 0) = \Pi_1(1, 0, \bar{\sigma}_1)$, $\underline{\sigma}_2$ by $\Pi_2(1, 1) = \Pi_2(1, 0, \underline{\sigma}_2)$ and $\bar{\sigma}_2$ by $\Pi_2(0, 0) = \Pi_2(1, 0, \bar{\sigma}_2)$. Then, we obtain the following result:

Proposition 3 (*Existence of equilibria with asymmetric firms*)

Compared to the case of symmetric firms:

- (i) The regions, in which both firms innovate or none of them innovates (Regions I and III) become smaller when firms are asymmetric, as for given values of d we have $\underline{\sigma}_2 < \underline{\sigma}$ and $\bar{\sigma}_1 > \bar{\sigma}$.
- (ii) The region with multiple equilibria in which one of the firms innovates and the other imitates (Region II) becomes smaller, as for given d we have $\underline{\sigma}_1 > \underline{\sigma}$ and $\bar{\sigma}_2 < \bar{\sigma}$.
- (iii) A new region with a unique pure strategy SPNE emerges (Region IV). In this region the large firm is an innovator and the small firm an imitator.

The four regions with the resulting equilibria are displayed in Figure 3. A specific feature of this extension is that allowing for initially asymmetric firms leads to the emergence of an area where the larger firm is an innovator and the smaller firm chooses to imitate. Thus, an increase in a firm's relative dominance raises incentives for that firm to innovate and decreases those of the rival. Empirical evidence widely supports this result. For instance, Henderson and Cockburn (1996) using data from individual research programs of pharmaceutical firms in the United States, suggest the advantage of large firms in the conduct of basic research.

The difference between the equilibria obtained in Region II and Region IV can be explained in terms of the persistence of firms' R&D strategies. In Region II we have equilibria where the optimal strategy of a firm is opposed to that of the rival. If the rival innovates the best reply is to imitate, and vice versa. Therefore, in a repeated context of this game, firms will not follow a continuous innovation strategy in Region II. On the contrary, in Region IV initially asymmetric firms always choose the same R&D strategy. The larger firm innovates and the smaller firm imitates. So, in Region IV, both firms continuously choose the same R&D strategy. The results in Proposition 3 allow us to obtain testable predictions of how market conditions such as product differentiation, firm asymmetries and spillovers affect firm's R&D strategy choice.

Proposition 4 (*The effect of ϵ , firm asymmetry, on aggregated output and social welfare*)

- (i) Aggregated industry output is constant in ϵ in all regions, and $q(0, 0) < q(1, 0, \sigma) = q(0, 1, \sigma) < q(1, 1)$.
- (ii) When changes in ϵ do not yield changes in firms' R&D strategies, in all region there exists at least one equilibrium in which social welfare is increasing in ϵ .
- (iii) When changes in ϵ yield changes in firms' R&D strategies, social welfare decreases when the number of innovators decreases.

Endogenizing a firm's decision to innovate or to abstain from innovation we obtain that asymmetries between firms, which may lead to a persistent innovator-imitator configuration in the market, can have both a positive and a negative effect on social welfare. A positive effect on welfare is obtained when greater dominance does not affect a firms' R&D decision. Then, the output increase of the dominant firm more than compensates the output loss by the smaller firm.⁹ Furthermore, a positive effect on social welfare is also obtained when market dominance means that the larger firm becomes an innovator. However, when more market dominance causes the smaller firm to abstain from innovation, the welfare effect is negative. This happens when products are highly differentiated and spillovers are low. Therefore, in situations in which firms are already protected against competitors market dominance is bad. We conclude that dominance is good for innovation when property rights are weak (spillovers are large) and competition is high but discourages innovation when the IPR protection is large and competition is weak.

3.2 n firms

In this section we analyze how the results extend to oligopoly markets with n initially symmetric firms. In this case, the corresponding inverse demand function of good i is given by:

$$p_i = a - bq_i - d \sum_{j \neq i} q_j, \quad i = 1, \dots, n, \quad (17)$$

We assume that spillovers occur when at least one firm decides to innovate. Thus, unit production cost are:

$$c_i(x_i, x_{-i}) = \begin{cases} c - \gamma & \text{if } x_i = 1 \\ c - \gamma\sigma & \text{if } x_i = 0 \text{ and } \exists j \text{ with } x_j = 1 \\ c & \text{else} \end{cases}, \quad i = 1, \dots, n. \quad (18)$$

In *stage 2*, firm i chooses the output q_i to maximize its operating profit:

$$\pi_i = \left(a - bq_i - d \sum_{j \neq i} q_j \right) q_i - c_i q_i, \quad i, j = 1, 2, i \neq j. \quad (19)$$

The Nash-Cournot equilibrium output for firm i is given by:

$$q_i = \frac{(2b - d)(a - c_i) - d(nc_i - \sum_{j=1}^n c_j)}{(2b - d)(2b + d(n - 1))}. \quad (20)$$

In *stage 1*, firms choose their profit-maximizing R&D strategy. When all firms engage in R&D (i.e. $x_1 = \dots = x_n = 1$), outputs and profits are:

$$q_i(1, \dots, 1) = \frac{a - c + \gamma}{2b + d(n - 1)} \quad \text{and} \quad \Pi_i(1, \dots, 1) = bq_i(1, \dots, 1)^2 - K. \quad (21)$$

If none of the firms engages in R&D (i.e. $x_1 = \dots = x_n = 0$), output and profits are equal to those of the classical Cournot model with differentiated products:

$$q_i(0, \dots, 0) = \frac{a - c}{2b - d + dn} \quad \text{and} \quad \Pi_i(0, \dots, 0) = bq_i(0, \dots, 0)^2. \quad (22)$$

⁹This result is similar to Tishler and Milstein (2009). For example, with $d = 0$ it means that it is better to have one monopolist in one large market than to have two monopolists in two small markets.

Furthermore, if all firms except one, say firm 1, engage in R&D the corresponding output and profit of firm 1 are given by:

$$q_1(0, 1, \dots, 1, \sigma) = \frac{(2b-d)(a-c+\sigma\gamma) - (n-1)(1-\sigma)\gamma d}{(2b-d)(2b+d(n-1))} \quad \text{and} \quad \Pi_1(0, 1, \dots, 1, \sigma) = bq_1(0, 1, \dots, 1, \sigma)^2. \quad (23)$$

Finally, if none of the firms innovates, except one, say firm 1, the corresponding output and profit of firm 1 are given by:

$$q_1(1, 0, \dots, 0, \sigma) = \frac{(2b-d)(a-c+\gamma) + (n-1)(1-\sigma)\gamma d}{(2b-d)(2b+d(n-1))} \quad \text{and} \quad \Pi_1(1, 0, \dots, 0, \sigma) = bq_1(1, 0, \dots, 0, \sigma)^2 - K. \quad (24)$$

To analyze how the frontiers of Region I and Region III depend on the number of firms in the market, we examine a firm's choice between innovation and imitation. First, we assume that all other firms in the industry innovate. Second, we assume that all other firms do not engage in R&D. Let $\underline{\sigma}_n$ be such that $\Pi_i(1, 1, \dots, 1) = \Pi_1(0, 1, \dots, 1, \underline{\sigma}_n)$ and $\bar{\sigma}_n$ such that $\Pi_1(1, 0, \dots, 0, \bar{\sigma}_n) = \Pi_i(0, 0, \dots, 0)$.

Proposition 5 (*The effect of n , competitive pressure, on equilibria*)

Compared to the duopoly case with two symmetric firms we have:

- (i) The region, in which all firms innovate (Region I) decreases with the number of firms in the market as for given d we have $\underline{\sigma}_n < \underline{\sigma}_{n-1}$.
- (ii) The region, in which none of the firms innovates (Region III) increases with the number of firms in the market as for given d we have $\bar{\sigma}_n < \bar{\sigma}_{n-1}$.

Figure 4 displays how Regions I and III change when the number of firms in the market increases. Regarding Region I, we observe that the probability of a particular firm to engage in R&D decreases as the number of competitors increases. With more competitors initially symmetric firms will be innovators only when products are sufficiently differentiated and IPR protection is high. This finding is supported by empirical evidence from Shankar et al. (1998). Though Region I shrinks with entry, notice that the overall innovative performance in the market increases within Region I as entrants also engage in R&D. With more competitors what was formerly Region II becomes more complex as further possible equilibria emerge. For example, with three firms we can have multiple equilibria with one innovator and two imitators or with two innovators and one imitator. Concerning Region III, we find that the entry of new firms means that equilibria with no innovating firm will occur for lower spillover values and for more differentiated products. Together, these results imply that the effect of entry on total R&D performance and welfare depends on spillovers and product differentiation. Concretely, we get the following result.

Proposition 6 (*The effect of n , competitive pressure, on industry R&D output and welfare*)

- (i) Entry increases (decreases) total R&D output and welfare when spillovers are low and products are highly differentiated (spillovers are high and products are rather homogenous).
- (ii) For given parameter values of (d, σ) an increase in n can first increase R&D output and welfare and then decrease it.

The first result highlights the role of both, IPR protection and the degree of product differentiation to assess the effect of changes in competition. Generally, we can say that more competition is good in markets with highly differentiated products and when IPR protection is high. On the contrary, increased competition is bad when products are homogeneous and IPR protection is weak. The second result indicates how changes in competitive pressure (the number of firms in the market) affect innovative effort. Notice, that with entry we can have more innovators. However, the (d, σ) -spaces in which all firms innovate and in which no firm innovates increase with entry. For example, from Figure 4 we see for $(d, \sigma) = (0.15, 0.1)$ that when the number of firms in the market passes from 3 to 5 to 10, total innovative effort passes from 3 to 5 to 0. So, as in the Aghion et al. (2005) growth model we find an inverted U-shaped relationship between innovative effort and competitive pressure. However, for higher values of σ , from the beginning, entry can yield to a decline of innovative effort and welfare, a result also found in Tishler and Milstein (2009) or De Bondt et al. (1992) because with high spillovers more rivals lead to reduced investments, output and profitability and reduced social welfare. As a general result we find that the effect of changes in competitive pressure measured by the number of firms, again, depends on both IPR protection and the degree of product differentiation.

4 Concluding remarks

This paper analyzes how the equilibrium R&D strategies of firms are affected by external factors such as spillovers and competitive pressure. Our analysis contributes to the understanding of a firm's R&D strategy choice. In this paper, especially, we focus on a firms' choice to innovate or to imitate. From the model we obtain that when firms choose endogenously their optimal R&D strategies three types of equilibria arise: equilibria in which all firms innovate, equilibria in which firms choose asymmetric R&D strategies with one innovating and one imitating firm, and, finally, equilibria in which no firm innovates. We find that stronger intellectual property rights protection provides higher incentives for firms to engage in R&D. Nevertheless, smaller firms are less likely to be innovators in markets with homogenous product and high levels of spillovers. The welfare consequences of having a dominate firm can be positive or negative. This depends on whether market dominance discourages smaller firms to innovate. An increase in the number of competitors can first increase innovative effort and welfare but then decrease them. So, our model would explain an inverted U-shaped relationship between innovative effort and competitive pressure. However, this relationship will depend on both the degree of product differentiation and IPR protection. Regarding social welfare, if a market is characterized by a high rate of innovation a reduction of IPR protection can discourage innovative performance and welfare substantially. However, a reduction of IPR protection can also increase social welfare because it may induce imitation and resulting in higher aggregate industry output.

Some important policy implications are obtained from our results. We find that a common IPR protection policy irrespective specific market and firms characteristics is inappropriate. The analysis of spillover effects on social welfare shows that a reduction of IPR protection can discourage innovative performance but also allow for imitation with a positive total welfare effect. Another implication of our findings is that the IPR protection policy must be tightly coordinated with the competition policy because external parameters such as IPR protection and competitive pressure jointly affect the firms' R&D strategy choice. Naturally, the implementation of such a policy is not an easy task because sectors might not be easily identified or firms might not be easily assigned to a specific sector. However, because the welfare gains from sector specific R&D policies might be substantial, future research should help us to identify these sectors and to indicate the appropriate R&D policy for them.

5 Appendix.

5.1 Proof of Proposition 1

Define $\bar{\sigma}$ such that firm i is indifferent between engaging and abstaining from R&D when firm j imitates:

$$\Pi_1(1, 0, \bar{\sigma}) - \Pi_i(0, 0) = b \left(\frac{2b(a-c+\gamma) - d(a-c+\bar{\sigma}\gamma)}{4b^2 - d^2} \right)^2 - K - b \left(\frac{2b(a-c) - d(a-c)}{4b^2 - d^2} \right)^2 = 0, \quad (25)$$

that is:

$$\bar{\sigma} = \frac{2b(a-c+\gamma) - d(a-c)}{d\gamma} - \frac{(4b^2 - d^2)}{d\gamma} \sqrt{\frac{K}{b} + \left(\frac{a-c}{2b+d} \right)^2}. \quad (26)$$

Define $\underline{\sigma}$ such that firm i is indifferent between engaging and abstaining from R&D when firm j engages in R&D:

$$\Pi_i(1, 1) - \Pi_2(1, 0, \underline{\sigma}) = b \left(\frac{2b(a-c+\gamma) - d(a-c+\gamma)}{4b^2 - d^2} \right)^2 - K - b \left(\frac{2b(a-c+\underline{\sigma}\gamma) - d(a-c+\gamma)}{4b^2 - d^2} \right)^2 = 0 \quad (27)$$

that is:

$$\underline{\sigma} = \frac{-2b(a-c) + d(a-c+\gamma)}{2b\gamma} + \frac{(4b^2 - d^2)}{2b\gamma} \sqrt{\left(\frac{a-c+\gamma}{2b+d} \right)^2 - \frac{K}{b}} \quad (28)$$

First, consider the partial derivatives. From equations (26) and (28) we have: $\partial\bar{\sigma}/\partial K < 0$, $\partial\underline{\sigma}/\partial K < 0$, $\partial\bar{\sigma}/\partial\gamma = -\bar{\sigma}/\gamma < 0$, $\partial\underline{\sigma}/\partial\gamma = -\underline{\sigma}/\gamma < 0$,

$$\partial\bar{\sigma}/\partial(a-c) = \frac{2b-d}{d\gamma} \left[1 - \left(\frac{a-c}{2b+d} \right) \left(\frac{K}{b} + \left(\frac{a-c}{2b+d} \right)^2 \right)^{-1/2} \right] > 0, \text{ and} \quad (29)$$

$$\partial\underline{\sigma}/\partial(a-c) = -\frac{2b-d}{2b\gamma} \left[1 - \left(\frac{a-c+\gamma}{2b+d} \right) \left(\left(\frac{a-c+\gamma}{2b+d} \right)^2 - \frac{K}{b} \right)^{-1/2} \right] > 0. \quad (30)$$

To prove existence of the equilibria we make the following claims:

Claim 1: $\bar{\sigma} > \underline{\sigma}$.

We have:

$$\bar{\sigma} - \underline{\sigma} = \frac{(4b^2 - d^2)}{\gamma d} \left(\frac{(a-c+\gamma)}{2b} - \sqrt{\frac{K}{b} + \left(\frac{a-c}{2b+d} \right)^2} - \frac{d}{2b} \sqrt{\left(\frac{a-c+\gamma}{2b+d} \right)^2 - \frac{K}{b}} \right) \quad (31)$$

This is an increasing function in K under assumption 1, i.e.

$$\frac{\partial\bar{\sigma} - \underline{\sigma}}{\partial K} = \frac{(4b^2 - d^2)}{2b\gamma d} \left(-\left(\frac{K}{b} + \left(\frac{a-c}{2b+d} \right)^2 \right)^{-\frac{1}{2}} + \frac{d}{2b} \left(\left(\frac{a-c+\gamma}{2b+d} \right)^2 - \frac{K}{b} \right)^{-\frac{1}{2}} \right) > 0 \quad (32)$$

for $K > \underline{K}$. Therefore, a sufficient condition for $\bar{\sigma} > \underline{\sigma}$ is that the condition holds for $K = \underline{K}$:

$$\begin{aligned}\bar{\sigma} - \underline{\sigma} &= \frac{(4b^2 - d^2)}{\gamma d} \left(\frac{(a - c + \gamma)}{2b} - \sqrt{\frac{\underline{K}}{b} + \left(\frac{a - c}{2b + d} \right)^2} - \frac{d}{2b} \sqrt{\left(\frac{a - c + \gamma}{2b + d} \right)^2 - \frac{\underline{K}}{b}} \right) \\ &= \frac{1}{2b} (2b - d) > 0.\end{aligned}\tag{33}$$

Claim 2: $\bar{\sigma} < 1$.

From assumption 1 we have that $K > \underline{K} = b \left(\frac{a - c + \gamma}{2b + d} \right)^2 - b \left(\frac{a - c}{2b + d} \right)^2$. Thus

$$\bar{\sigma} < \frac{2b(a - c + \gamma) - d(a - c)}{d\gamma} - \frac{(4b^2 - d^2)}{d\gamma} \sqrt{\frac{\underline{K}}{b} + \left(\frac{a - c}{2b + d} \right)^2} = 1\tag{34}$$

Claim 3: $\underline{\sigma} > 0$.

From assumption 1 we have $K < \bar{K} = b \left(\frac{a - c + \gamma}{2b + d} \right)^2 - b \left(\frac{(2b - d)(a - c) - d\gamma}{4b^2 - d^2} \right)^2$. Thus, $\underline{\sigma} > \frac{-2b(a - c) + d(a - c + \gamma)}{2b\gamma} + \frac{(4b^2 - d^2)}{2b\gamma} \sqrt{\left(\frac{a - c + \gamma}{2b + d} \right)^2 - \frac{\bar{K}}{b}} = 0$.

Together, claims 1-3 prove the existence of the different equilibria.

5.2 Proof of Proposition 2

First, consider aggregated output. We have:

$$q(1, 1) = \frac{2(a - c + \gamma)}{2b + d} > q(1, 0, \sigma) = \frac{2(a - c) + (1 + \sigma)\gamma}{2b + d} > q(0, 0) = \frac{2(a - c)}{2b + d}\tag{35}$$

and $\partial q(1, 0, \sigma)/\partial \sigma > 0$ which proves the statements regarding aggregated output.

Next, consider social welfare. When both firms engage in R&D, (i.e. $x_i = 1$, $i = 1, 2$) social welfare is:

$$W(1, 1) = (3b + d) \left(\frac{a - c + \gamma}{2b + d} \right)^2 - 2K\tag{36}$$

If none of the firms engages in R&D, (i.e. $x_i = 0$, $i = 1, 2$) social welfare is:

$$W(0, 0) = (3b + d) \left(\frac{a - c}{2b + d} \right)^2\tag{37}$$

Finally, if firm 1 engages in R&D and firm 2 decides to imitate, social welfare is:

$$\begin{aligned}W(1, 0, \sigma) &= \Pi_1(1, 0, \sigma) + \Pi_2(0, 1, \sigma) + dq_1(1, 0, \sigma)q_2(0, 1, \sigma) + \frac{1}{2}b(q_1^2(1, 0, \sigma) + q_2^2(0, 1, \sigma)) \\ &= \frac{3}{2}b \left(\frac{a - c}{2b + d} + \frac{(2b - d\sigma)\gamma}{4b^2 - d^2} \right)^2 + \frac{3}{2}b \left(\frac{a - c}{2b + d} + \frac{(2b\sigma - d)\gamma}{4b^2 - d^2} \right)^2 \\ &\quad + d \left(\frac{a - c}{2b + d} + \frac{(2b - d\sigma)\gamma}{4b^2 - d^2} \right) \left(\frac{a - c}{2b + d} + \frac{(2b\sigma - d)\gamma}{4b^2 - d^2} \right) - K\end{aligned}\tag{38}$$

To prove statement (i), from (36) and (37) we have:

$$\begin{aligned}
W(1, 1) - W(0, 0) &= (2b + d)^{-2} (2a - 2c + \gamma) (3b + d) \gamma - 2K \\
&> (2b + d)^{-2} (2a - 2c + \gamma) (3b + d) \gamma - 2\bar{K} \\
&= \frac{2(b - d)(4b^2 - d^2)(a - c) + (4b^3 - bd^2 + d^3)\gamma}{(4b^2 - d^2)^2} \gamma > 0
\end{aligned} \tag{39}$$

To prove statement (ii), consider the second derivative of (38):

$$\frac{\partial^2 W(1, 0, \sigma)}{\partial \sigma^2} = \frac{(12b^2 - d^2)b\gamma^2}{(4b^2 - d^2)^2} > 0. \tag{40}$$

Finally, to prove statement (iii), we analyze when

$$W(1, 0, \underline{\sigma}) < W(1, 1). \tag{41}$$

By definition of $\underline{\sigma}$ we have $\Pi_2(0, 1, \underline{\sigma}) = \Pi_2(1, 1)$. So, (41) is equivalent to

$$\Pi_1(1, 0, \underline{\sigma}) + dq_1(1, 0, \underline{\sigma})q_2(0, 1, \underline{\sigma}) + \frac{1}{2}b(q_1^2(1, 0, \underline{\sigma}) + q_2^2(0, 1, \underline{\sigma})) < \Pi_1(1, 1) + (d + b)q_1(1, 1)q_2(1, 1) \tag{42}$$

or

$$\frac{1}{2}\gamma(\underline{\sigma} - 1)(2(b - d)(a - c) + \gamma(b - 2d + b\underline{\sigma})) < 0 \tag{43}$$

or

$$\frac{1}{4}\gamma(\underline{\sigma} - 1) \left((2b - 3d)(a - c + \gamma) + (4b^2 - d^2) \sqrt{\left(\frac{a - c + \gamma}{2b + d} \right)^2 - \frac{K}{b}} \right) < 0. \tag{44}$$

This is true if $2b > 3d$ or $K < \frac{8(b-d)(a-c+\gamma)^2bd}{(4b^2-d^2)^2}$. Notice, that in case of homogeneous products the conditions are not fulfilled such that $W(1, 0, \underline{\sigma}) > W(1, 1)$.

Next, we analyze when

$$W(1, 0, \bar{\sigma}) > W(0, 0). \tag{45}$$

By definition of $\bar{\sigma}$ we have $\Pi_1(1, 0, \bar{\sigma}) = \Pi_1(0, 0)$. So (45) is equivalent to

$$\Pi_2(0, 1, \bar{\sigma}) + dq_1(1, 0, \bar{\sigma})q_2(0, 1, \bar{\sigma}) + \frac{1}{2}b(q_1^2(1, 0, \bar{\sigma}) + q_2^2(0, 1, \bar{\sigma})) > \Pi_2(0, 0) + (d + b)q_1^2(0, 0) \tag{46}$$

or

$$2(a - c)(b - d + (3b - d)\underline{\sigma}) + \gamma(b + 3b\sigma^2 - 2d\underline{\sigma}) > 0 \tag{47}$$

which always holds.

5.3 Proof of Proposition 3

From the definition of $\underline{\sigma}_1$, $\bar{\sigma}_1$, $\underline{\sigma}_2$ and $\bar{\sigma}_2$ we obtain:

$$\underline{\sigma}_1 = -\frac{2b\epsilon M - d(1-\epsilon)M - d\gamma}{2b\gamma} + \frac{4b^2 - d^2}{2b\gamma} \sqrt{\frac{(2b\epsilon M - d(1-\epsilon)M + \gamma(2b-d))^2}{(4b^2 - d^2)^2} - \frac{K}{b}} \quad (48)$$

$$\bar{\sigma}_1 = \frac{2b(M\epsilon + \gamma) - Md(1-\epsilon)}{d\gamma} - \frac{(4b^2 - d^2)}{d\gamma} \sqrt{\frac{K}{b} + \frac{(2b\epsilon M - d(1-\epsilon)M)^2}{(4b^2 - d^2)^2}} \quad (49)$$

$$\underline{\sigma}_2 = -\frac{2b(1-\epsilon)M - d(\epsilon M + \gamma)}{2b\gamma} + \frac{(4b^2 - d^2)}{2b\gamma} \sqrt{\frac{(2b(1-\epsilon)M - d\epsilon M + \gamma(2b-d))^2}{(4b^2 - d^2)^2} - \frac{K}{b}} \quad (50)$$

$$\bar{\sigma}_2 = \frac{2b(1-\epsilon)M - d\epsilon M + 2b\gamma}{d\gamma} - \frac{4b^2 - d^2}{d\gamma} \sqrt{\left(\frac{2b(1-\epsilon)M - d\epsilon M}{4b^2 - d^2}\right)^2 + \frac{K}{b}} \quad (51)$$

Because $\epsilon \in (\frac{1}{2}, 1)$, statement (i) is true if

$$\frac{\partial \underline{\sigma}_2}{\partial \epsilon} = \frac{(2b+d)M}{2b\gamma} \left(1 - \frac{\frac{(2b(1-\epsilon)M - d\epsilon M + \gamma(2b-d))}{(4b^2 - d^2)}}{\sqrt{\frac{(2b(1-\epsilon)M - d\epsilon M + \gamma(2b-d))^2}{(4b^2 - d^2)^2} - \frac{K}{b}}} \right) < 0 \text{ and} \quad (52)$$

$$\frac{\partial \bar{\sigma}_1}{\partial \epsilon} = \frac{(2b+d)M}{d\gamma} \left(1 - \frac{\frac{(2b\epsilon M - d(1-\epsilon)M)}{(4b^2 - d^2)}}{\sqrt{\frac{K}{b} + \frac{(2b\epsilon M - d(1-\epsilon)M)^2}{(4b^2 - d^2)^2}}} \right) > 0 \quad (53)$$

which holds if $K > 0$.

Similarly, statement (ii) is true if

$$\frac{\partial \underline{\sigma}_1}{\partial \epsilon} = \frac{(2b+d)M}{2b\gamma} \left(-1 + \frac{\frac{2(2b\epsilon M - d(1-\epsilon)M + \gamma(2b-d))}{(4b^2 - d^2)}}{\sqrt{\frac{(2b\epsilon M - d(1-\epsilon)M + \gamma(2b-d))^2}{(4b^2 - d^2)^2} - \frac{K}{b}}} \right) > 0 \text{ and} \quad (54)$$

$$\frac{\partial \bar{\sigma}_2}{\partial \epsilon} = \frac{(2b+d)M}{d\gamma} \left(-1 + \frac{\frac{2(2b(1-\epsilon)M - d\epsilon M)}{(4b^2 - d^2)}}{\sqrt{\left(\frac{2b(1-\epsilon)M - d\epsilon M}{4b^2 - d^2}\right)^2 + \frac{K}{b}}} \right) < 0 \quad (55)$$

which also holds if $K > 0$.

Finally, statement (iii) follows directly from the former two. When all regions shrink, a new region must emerge. The characteristics of the equilibrium in this region follow from the definition of the regions' frontiers.

5.4 Proof of Proposition 4

Statement (i) follows immediately from:

$$q(0,0) = \frac{M}{2b+d} < q(1,0,\sigma) = q(0,1,\sigma) = \frac{M + (1+\sigma)\gamma}{2b+d} < q(1,1) = \frac{M + 2\gamma}{2b+d} \quad (56)$$

which are all independent from ϵ .

To prove statement (ii), consider the social welfare in the different regions:

$$W(0,0) = \frac{(12b^2 - d^2)b - 2\epsilon(1-\epsilon)(3b-d)(2b+d)^2}{2(4b^2 - d^2)^2} M^2 \quad (57)$$

$$W(1,1) = \frac{2\gamma(3b+d)(2b-d)^2(M+\gamma) + \left((12b^2 - d^2)b - 2\epsilon(1-\epsilon)(3b-d)(2b+d)^2\right) M^2}{2(4b^2 - d^2)^2} - 2K \quad (58)$$

$$W(1,0,\sigma) = \frac{3}{2}b \left(\frac{2b\epsilon M - d(1-\epsilon)M + (2b-d\sigma)\gamma}{4b^2 - d^2} \right)^2 + \frac{3}{2}b \left(\frac{2b(1-\epsilon)M - d\epsilon M + (2b\sigma-d)\gamma}{4b^2 - d^2} \right)^2 + d \frac{2b\epsilon M - d(1-\epsilon)M + (2b-d\sigma)\gamma}{4b^2 - d^2} \frac{2b(1-\epsilon)M - d\epsilon M + (2b\sigma-d)\gamma}{4b^2 - d^2} - K \quad (59)$$

$$W(0,1,\sigma) = \frac{3}{2}b \left(\frac{2b\epsilon M - d(1-\epsilon)M + (2b\sigma-d)\gamma}{4b^2 - d^2} \right)^2 + \frac{3}{2}b \left(\frac{2b(1-\epsilon)M - d\epsilon M + (2b-d\sigma)\gamma}{4b^2 - d^2} \right)^2 + d \frac{2b\epsilon M - d(1-\epsilon)M + (2b\sigma-d)\gamma}{4b^2 - d^2} \frac{2b(1-\epsilon)M - d\epsilon M + (2b-d\sigma)\gamma}{4b^2 - d^2} - K \quad (60)$$

Differentiation with respect to ϵ yields:

$$\frac{\partial W(0,0)}{\partial \epsilon} = \frac{\partial W(1,1)}{\partial \epsilon} = \frac{(2\epsilon-1)(3b-d)(2b+d)^2}{(4b^2 - d^2)^2} M^2 \geq 0 \quad (61)$$

$$\frac{\partial W(1,0,\sigma)}{\partial \epsilon} = \frac{((2\epsilon-1)M + \gamma(1-\sigma))(3b-d)M}{(2b-d)^2} \geq 0 \quad (62)$$

for $\epsilon \in (\frac{1}{2}, 1)$. This guarantees that social welfare increases with ϵ in Regions I, III and IV in which we have a unique equilibrium. Furthermore, the last expression is sufficient to guarantee that there is at least one equilibrium in Region II in which social welfare increases with ϵ . This happens when the large firm is the innovator and the small firm the imitator. In the opposite case, in which the small firm is the innovator and the large firm the imitator we get

$$\frac{\partial W(0,1,\sigma)}{\partial \epsilon} = \frac{((2\epsilon-1)M - \gamma(1-\sigma))}{(2b-d)^2} (3b-d)M. \quad (63)$$

Then, social welfare does not necessarily increase with ϵ .

To prove statement (iii), from Proposition 3 we know that a change in ϵ increases region IV. This can yield three changes in the firms' equilibrium R&D strategies. First, instead of two innovators we can have one innovator and one imitator. Notice that social welfare can be written as

$$W = \frac{3}{2}b \left[(q - q_2)^2 + q_2^2 \right] + d(q - q_2)q_2, \quad (64)$$

where $\partial W / \partial q > 0$ and $\partial W / \partial q_2 = -(3b-d)(q - 2q_2) < 0$ for $q_2 < q_1 = q - q_2$. Therefore, $q(1,1) > q(1,0,\sigma)$, $q_2(1,1) - q_2(1,0,\sigma) = 2b\gamma(1-\sigma) / (4b^2 - d^2) > 0$, $q_2(1,1) < q_1(1,1)$ and $q_2(1,0,\sigma) < q_1(1,0,\sigma)$ means that $W(1,1) - W(1,0,\sigma) > 0$, i.e. welfare is larger when both firms innovate than when firm 1 innovates and firm 2 imitates. Second, instead of a small innovator and a large imitator we can have a large innovator and a small imitates.

imitator. Then we have:

$$W(1, 0, \sigma) - W(0, 1, \sigma) = M\gamma(2\epsilon - 1)(1 - \sigma) \frac{3b - d}{(2b - d)^2} > 0 \quad (65)$$

Finally, we can pass from a situation with no innovator to one with a large innovator and a small imitator. Notice that social welfare can be written as

$$W = \frac{3}{2}b \left[q_1^2 + (q - q_1)^2 \right] + dq_1(q - q_1), \quad (66)$$

where $\partial W / \partial q > 0$ and $\partial W / \partial q_1 = -(3b - d)(q - 2q_1) > 0$ for $q_1 > q_2 = q - q_1$. Therefore, $q(0, 0) < q(1, 0, \sigma)$, $q_1(0, 0) - q_1(1, 0, \sigma) = -(2b - d\sigma)\gamma / (4b^2 - d^2) < 0$, $q_1(0, 0) > q_2(0, 0)$ and $q_1(1, 0, \sigma) > q_2(1, 0, \sigma)$ means that $W(1, 0, \sigma) - W(0, 0) > 0$, i.e. welfare is larger when firm 1 innovates and firm 2 imitates than when no firm innovates.

5.5 Proof of Proposition 5

To prove statement (i), from the definition of $\underline{\sigma}_n$ by $\Pi_i(1, \dots, 1) - \Pi_1(0, 1, \dots, 1, \underline{\sigma}_n) = 0$ we get:

$$\begin{aligned} \underline{\sigma}_n &= \frac{-(2b - d)(a - c) + (n - 1)\gamma d}{\gamma(2b - 2d + dn)} + \frac{(2b - d)(2b + d(n - 1))}{\gamma(2b - 2d + dn)} \sqrt{\left(\frac{a - c + \gamma}{2b - d + dn}\right)^2 - \frac{K}{b}} \\ &= \frac{-(2b - d)(a - c + \gamma)}{\gamma(2b - 2d + dn)} + 1 + \left(\frac{(2b - d)}{\gamma} + \frac{(2b - d)d}{\gamma(2b - 2d + dn)}\right) \sqrt{\left(\frac{a - c + \gamma}{2b - d + dn}\right)^2 - \frac{K}{b}} \end{aligned} \quad (67)$$

From differentiation we get:

$$\begin{aligned} \frac{\partial \underline{\sigma}_n}{\partial n} &= \frac{d(2b - d)(a - c + \gamma)}{\gamma(2b - 2d + dn)^2} - d \left(\frac{(2b - d)d}{\gamma(2b - 2d + dn)^2} \right) \sqrt{\left(\frac{a - c + \gamma}{2b - d + dn}\right)^2 - \frac{K}{b}} \\ &\quad - d \left(\frac{(2b - d)}{\gamma} + \frac{(2b - d)d}{\gamma(2b - 2d + dn)} \right) \left(\left(\frac{a - c + \gamma}{2b - d + dn}\right)^2 - \frac{K}{b} \right)^{-\frac{1}{2}} \left(\frac{a - c + \gamma}{2b - d + dn}\right)^2 (2b - d + dn)^{-1} \\ &< 0 \end{aligned} \quad (68)$$

iff

$$\frac{K}{b} + \frac{(2b - 3d + dn)(a - c + \gamma)^2}{(2b - d + dn)d^2} > 0 \quad (69)$$

which always holds.

To prove statement (ii), from the definition of $\bar{\sigma}_n$ by $\Pi_1(1, 0, \dots, 0, \bar{\sigma}_n) - \Pi_i(0, 0, \dots, 0) = 0$ we get:

$$\bar{\sigma}_n = 1 + \frac{(2b - d)(a - c + \gamma)}{(n - 1)\gamma d} - \frac{(2b - d)(2b + d(n - 1))}{(n - 1)\gamma d} \sqrt{\left(\frac{a - c}{2b - d + dn}\right)^2 + \frac{K}{b}} \quad (70)$$

From differentiation we get:

$$\begin{aligned} \frac{\partial \bar{\sigma}_n}{\partial n} &= -\frac{(2b-d)(a-c+\gamma)}{(n-1)^2 \gamma d} + \frac{2(2b-d)b}{(n-1)^2 d \gamma} \sqrt{\left(\frac{a-c}{2b-d+dn}\right)^2 + \frac{K}{b}} \\ &\quad + d \frac{(2b-d)(2b+d(n-1))}{(n-1) \gamma d} \left(\left(\frac{a-c}{2b-d+dn}\right)^2 + \frac{K}{b} \right)^{-\frac{1}{2}} \left(\frac{a-c}{2b-d+dn}\right)^2 (2b-d+dn)^{-1} \\ &< 0 \end{aligned} \quad (71)$$

iff

$$-(a-c+\gamma) + 2b \sqrt{\left(\frac{a-c}{2b-d+dn}\right)^2 + \frac{K}{b}} + (n-1)d \left(\left(\frac{a-c}{2b-d+dn}\right)^2 + \frac{K}{b} \right)^{-\frac{1}{2}} \left(\frac{a-c}{2b-d+dn}\right)^2 < 0 \quad (72)$$

which is an increasing function in K . Therefore, a sufficient condition is that this holds for $\bar{K}$:

$$\bar{K} = b \left(\frac{a-c+\gamma}{2b-d+dn} \right)^2 - b \left(\frac{(2b-d)(a-c)-d\gamma}{(2b-d)(2b+d(n-1))} \right)^2 \quad (73)$$

defined by $\Pi_j(1, 1, \dots, 1) = \Pi_j(1, 0, \dots, 0, \sigma = 0)$, $j \neq 1$. This yields:

$$\begin{aligned} &-(a-c+\gamma) + 2b \sqrt{\left(\frac{a-c}{2b-d+dn}\right)^2 + \left(\frac{a-c+\gamma}{2b-d+dn}\right)^2 - \left(\frac{(2b-d)(a-c)-d\gamma}{(2b-d)(2b+d(n-1))}\right)^2} \\ &+ (n-1)d \left(\left(\frac{a-c}{2b-d+dn}\right)^2 + \left(\frac{a-c+\gamma}{2b-d+dn}\right)^2 - \left(\frac{(2b-d)(a-c)-d\gamma}{(2b-d)(2b+d(n-1))}\right)^2 \right)^{-\frac{1}{2}} \left(\frac{a-c}{2b-d+dn}\right)^2 \\ &< -\frac{(2(a-c)+\gamma)(n-1)d\gamma}{(a-c+\gamma)(2b-d+dn)} < 0 \end{aligned} \quad (74)$$

References

- [1] Aghion, P., Howitt, P. (1992). "A Model of Growth through Creative Destruction", *Econometrica*, 60(2), pp.323-351.
- [2] Aghion, P., Harris, C., Howitt, P., Vickers, J. (2001). "Competition, Imitation, and Growth with Step-by-Step Innovation", *Review of Economic Studies*, 68, pp.467-492.
- [3] Aghion, P., Bloom, N., Blundell, R., Griffith, R., Howitt, P. (2005). "Competition and innovation: an inverted-U relationship", *Quarterly Journal of Economics*, 120, pp.701-728.
- [4] Amir, R., Wooders, J. (2000). "One-Way Spillovers, Endogenous Innovator/Imitator Roles, and Research Joint Ventures", *Games and Economic Behavior*, 31, pp.1-25.
- [5] Arora, A., Gambardella A. (1994). "The Changing Technology of Technological Change: General and Abstract Knowledge and the Division of Innovative Labour", *Research Policy*, 32, pp. 523-532.

- [6] Arrow, K. (1962). "Economic welfare and the allocation of resources for invention", In: Nelson, R. (Ed.), *The Rate and Direction of Inventive Activity: Economic and Social Factors*. Princeton University Press, Princeton, NJ.
- [7] d'Aspremont C., Jaquemin, A. (1988). "Cooperative and noncooperative R&D in duopoly with spillovers", *American Economic Review*, 78 (5), pp.1133-1137.
- [8] Bessen, J., Maskin, E. (2009). "Sequential Innovation, Patents, and Imitation", *RAND Journal of Economics*, 40, pp.611-635.
- [9] Blundell, R., Griffith, R., Van Reenen, J. (1999). "Market Share, Market Value and Innovation in a Panel of British Manufacturing Firms", *Review of Economic Studies*, 66, pp.529-554.
- [10] Boldrin, M., Levine, D. (2009). "Market Size And Intellectual Property Protection," *International Economic Review*, 50(3), pp.855-881.
- [11] Bower, J. L., Christensen, C.M. (1995). "Disruptive technologies: catching the wave", *Harvard Business Review*, pp. 43-53.
- [12] Braguinsky, S., Gabdrakhmanov, S., Ohyama, A. (2007). "A theory of competitive industry dynamics with innovation and imitation", *Review of Economic Dynamics*, 10(4), pp.729-760.
- [13] Che, J., Qiu, L., Zhou, W. (2009). "Intellectual property rights enforcement in imperfect markets", Levine's Working Paper Archive, David K. Levine.
- [14] Cefis, E. (2003). "Is there persistence in innovative activities?," *International Journal of Industrial Organization*, 21(4), pp.489-515.
- [15] Cefis, E., Orsenigo, L. (2001). "The persistence of innovative activities. A cross-countries and cross-sectors comparative analysis", *Research Policy*, 30, pp.1139-1158.
- [16] Czarnitzki, D., Etro, F., Kraft, K. (2008). "The Effect of Entry on R&D Investment of Leaders: Theory and Empirical Evidence", ZEW Discussion Paper No. 08-078, Mannheim.
- [17] Dasgupta, P., Stiglitz, J. (1980). "Industrial Structure and the Nature of Innovative Activity", *Economic Journal*, 90, pp.226-293.
- [18] De Bondt, R., Slaets, P., Cassiman, B. (1992). "The degree of spillovers and the number of rivals for maximum effective R &D", *International Journal of Industrial Organization*, 10(1), pp.35-54.
- [19] De Bondt, R. (1996). "Spillovers and innovative activities", *International Journal of Industrial Organization*, 15(1), pp.1 - 28.
- [20] Fershtman, C., Markovich, S. (2010). "Patents, imitation and licensing in an asymmetric dynamic R&D race", *International Journal of Industrial Organization*, 28, pp.113-126.
- [21] Futia, C., (1980). "Schumpeterian Competition", *Quarterly Journal of Economics*, 94 (4), pp. 675-695.
- [22] Gans, J., Hsu, D.H., Stern, S. (2008). "The Impact of Uncertain Intellectual Property Rights on the Market for Ideas: Evidence for Patent Grant Delays", *Management Science*, 54 (5), pp.982-997.

- [23] Gans, J., Stern, S. (2003). "The product Market and the Market for Ideas: Commercialization Strategies for Technology Entrepreneurs", *Research Policy*, 32, pp.333-350.
- [24] Gallini, N.T., Scotchmer, S. (2002). "Intellectual property: When is it the best incentive system?", in: Jaffe, A., Lerner, J., Stern, S. (Eds.), *Innovation Policy and the Economy*, vol.2 MIT Press, Cambridge, MA.
- [25] Geroski P. (1990). "Innovation, Technological Opportunity, and Market Structure", *Oxford Economic Papers*, 42 (3), pp.586-602.
- [26] Gilbert, R., Newbery, D. (1982). "Preemptive patenting and the persistence of monopoly", *American Economic Review*, 72, pp.514-526.
- [27] Grossman, G., Helpman, E. (1991). "Endogenous Product Cycles", *The Economic Journal*, 101, pp.1214-1229.
- [28] Halmenschlager, C. (2006). "Spillovers And Absorptive Capacity In A Patent Race," Manchester School, University of Manchester, 74 (1), pp. 85-102, 01.
- [29] Helpman, E. (1993). "Innovation, Imitation, and Intellectual Property Rights," *Econometrica*, 61(6), pp.1247-1280.
- [30] Henderson, R., Cockburn, I. (1996). "Scale, Scope, and Spillovers: Determinants of Research Productivity in the Pharmaceutical Industry", *RAND Journal of Economics*, 27(1), pp.32-59.
- [31] Reinganum, J. (1983). "Uncertain innovation and the persistence of monopoly", *American Economic Review* 73 (4), pp.741-748.
- [32] Segestrom, P. (1991). "Innovation, Imitation, and Economic Growth," *Journal of Political Economy*, 99(4), pp. 807-27.
- [33] Shankar, V., Carpenter, G.S., Krishnamurthi, L. (1998). "Late Mover Advantage: How Innovative Late Entrants Outsell Pioneers", *Journal of Marketing Research*, 35, pp.54-70.
- [34] Schumpeter, J. (1934). "The theory of economic development: An inquiry into profits, capital, credit, interest and the business cycle" (transl. Theorie der wirtschaftlichen Entwicklung, 1911).
- [35] Schumpeter, J. (1942). Capitalism, Socialism, and Democracy. Allen & Unwin, London.
- [36] Theilen, B. (2012). "Product Differentiation and Competitive Pressure", *Journal of Economics*, DOI 10.1007/s00712-011-0261-5.
- [37] Tishler, A., Milstein, I. (2009). "R&D wars and the effects of innovation on the success and survivability of firms in oligopoly markets", *International Journal of Industrial Organization*, 27, pp.519-531.
- [38] Vives, X. (2008). "Innovation and Competitive Pressure", *The Journal of Industrial Economics* 56(3), pp.419-469.
- [39] Zhou, W. (2009). "Innovation, Imitation and Competition", *The B.E. Journal of Economic Analysis and Policy*, 9(1), article 27.

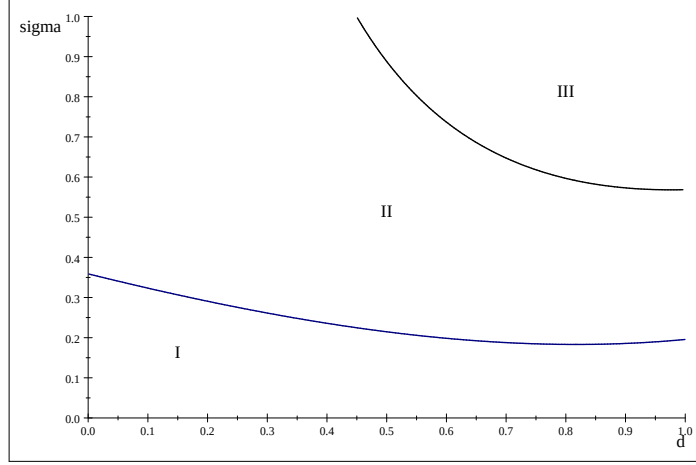
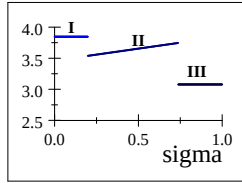


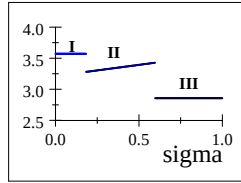
Figure 1: The three regions of model equilibria for $b = 1$, $\gamma = 1$, $a - c = 4$ and $K = 1, 5$.

Aggregated industry output

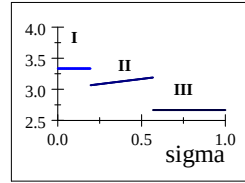
$d = 0.6$



$d = 0.8$

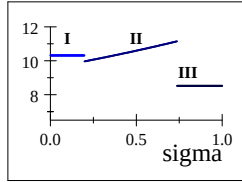


$d = 1$

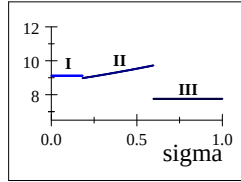


Social welfare

$d = 0.6$



$d = 0.8$



$d = 1$

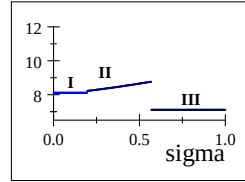


Figure 2: Aggregated industry output and social welfare as functions of σ for $b = 1$, $a - c = 4$, $\gamma = 1$, and $K = 1.5$.

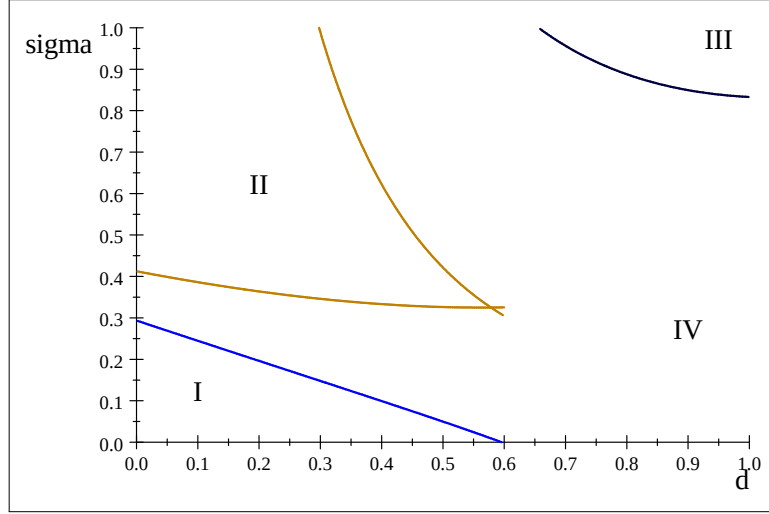


Figure 3: The four regions of model equilibria for $b = 1$, $\gamma = 1$, $M = 8$, $\epsilon = \frac{11}{20}$ and $K = 1.5$.

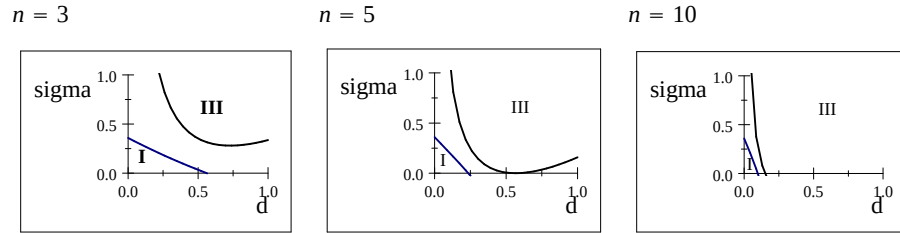


Figure 4: The regions of model equilibria for $b = 1$, $\gamma = 1$, $a - c = 4$ and $K = 1.5$.